

Airbnb Seattle Data Analysis Project



Submitted by: HARLEEN KAUR

Submitted to: Dr. Junaid Qazi

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Introduction

The primary goal of this project is to use Power BI to analyze Airbnb listings and bookings in Seattle from 2016 to 2017. Rich information are available from Airbnb, a global network that links visitors with hosts offering distinctive accommodations. These databases provide insights into property kinds, prices, bookings, and consumer behavior. Our objective is to provide an interactive dashboard that reveals important patterns and trends, like how pricing affects bookings, where listings are located, and how well hosts perform. We hope to provide insightful analysis of this data to decision-makers and hosts in the short-term rental industry.

Open-Source Data of Interest

The Airbnb Seattle Dataset, which is openly accessible through Kaggle, is the source of the data used in this research. This dataset includes all of Seattle's Airbnb listings and bookings for 2016–2017.

There are two main CSV files containing the data:

1. **Listings CSV:** Provides comprehensive details regarding every property, such as:
 - **Property ID:** A special number assigned to every listing.
 - **Nightly Rate:** The listing's cost per night.
 - **Host Information:** Specifics about the host, including name, host ID, and superhost status.
 - **Details about the property:** how many bedrooms, baths, and facilities it has, as well as if it is a separate room or the entire house.
2. **Booking CSV:** Provides comprehensive booking data, such as:
 - **Dates of Booking:** The dates of arrival and departure.
 - **Occupancy Status:** Details on whether or not a listing was reserved.
 - **Cancellation status:** Booking behavior can be better understood by knowing the Cancellation Status, which indicates if a reservation was canceled.

This open-source dataset is a great resource for learning how Airbnb listings function in a big city like Seattle because it covers the topic thoroughly and is applicable to actual Airbnb operations.

Purpose and Goals

This project's main goal is to use Power BI to analyze the Airbnb Seattle dataset and find important market insights, with a particular emphasis on pricing tactics, booking trends, and geography considerations. Our objectives are to:

- ✓ **Assist Hosts in Optimizing Listings:**
Our objective is to furnish hosts with data-driven insights to optimize their occupancy rates and modify pricing strategies in response to demand by scrutinizing price trends.
- ✓ **Give Travelers Insights:**
By using our research to determine the ideal neighborhoods and seasons to book, travelers can get better deals.
- ✓ **Investigate Neighborhood Dynamics:**
Knowing the geographic distribution of listings can help you determine which parts of Seattle are more in demand for short-term rentals as well as how demand and prices vary by neighborhood.

Key Questions

Our analysis was guided by the following primary questions:

Q1. What are the typical nightly prices for various kinds of properties, and how do these prices affect the way reservations are made?

Q2. Which months have the highest demand, and what are the seasonal booking trends?

Q3. Which neighborhoods are most densely packed with ads and reservations, and how do prices differ between these areas?

Why did we choose this dataset?

- **Valuable Insights:** The dataset provides a complete view of Seattle's Airbnb market, including listings, pricing, and booking trends.
- **Real-world dynamics:** The data reflects real-world dynamics in a major city's short-term rental market, offering valuable Findings for Executives and policymakers.

- **Variety Of Metrics:** It contains information on property types, locations, reviews, and availability, enabling in-depth analysis.
- **Visualization Potential:** The dataset is well-Structured for building an interactive dashboard in Power BI to Investigate patterns and trends.
- **Open Source:** Accessible through Kaggle, making it a well-documented and reliable source for analysis.

Metrics and KPIs

We established the following Key Performance Indicators (KPIs) and Metrics to monitor the efficacy of our analysis and provide answers to the important questions:

Average Price Per Night:

The mean cost per night for various property categories (e.g., individual room, complete home). This statistic aids in determining price fluctuations and how they affect reservations.

Rate of Occupancy:

the percentage of days available that are reserved for a property. While a lower rate might suggest overpricing or other problems, a greater occupancy rate shows that the hotel is more well-liked.

Volume of Monthly Bookings:

The total number of reservations made in a given month, which enables us to forecast demand all year round and identify peak booking periods.

Rate of Cancellations:

The proportion of canceled reservations relative to the total number of reservations, which aids in identifying peak cancellation season and seasonal patterns in cancellation rates.

Best Hosts:

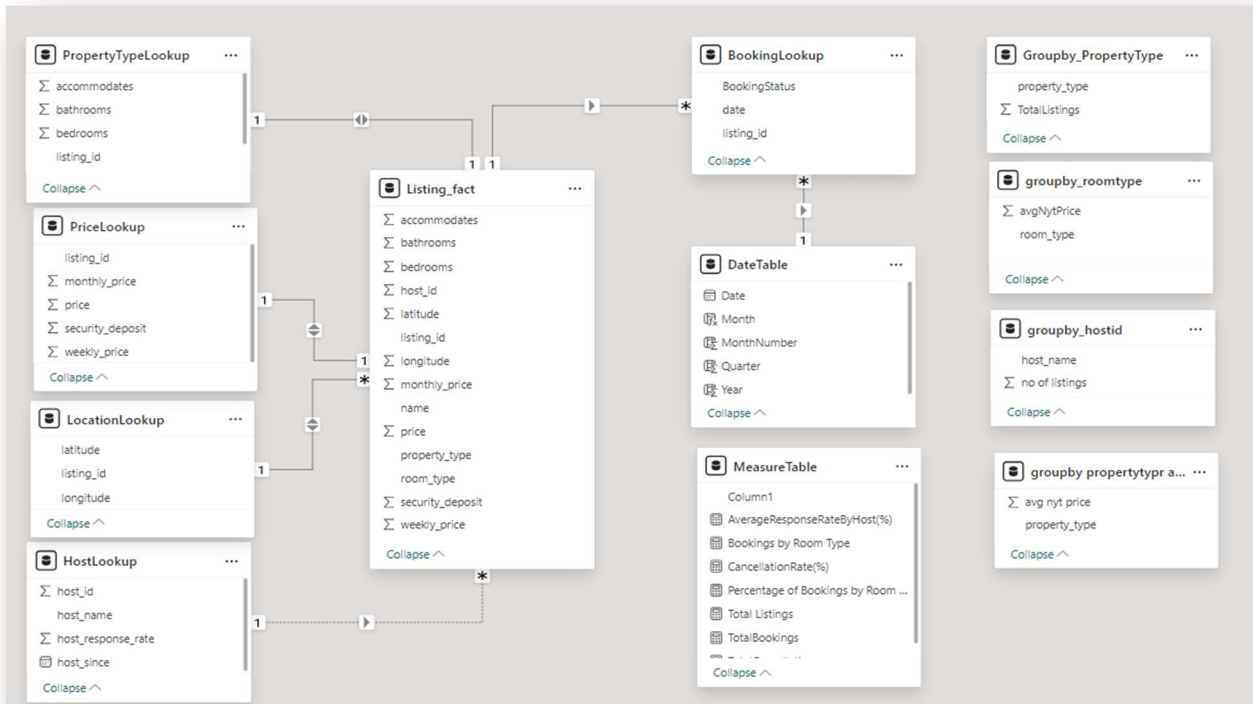
Finding the hosts who get the most bookings and the fewest cancellations. This enables us to identify industry best practices from hosts who have achieved success.

Density of Geographic Listings:

The quantity of listings and reservations in every neighborhood, which sheds light on the most popular places and the regional variations in pricing.

Data Model

The data model was developed by integrating the two key CSV files (Listings and Bookings) within Power BI.



RESULTS:

1. Total Listings:

One million listings make up the dataset, as indicated by the KPI card.

The total properties that are available within the chosen time frame are represented by this.



2. Total Bookings:



An additional KPI card showing 524K (524,000 bookings) as the overall number of reservations.

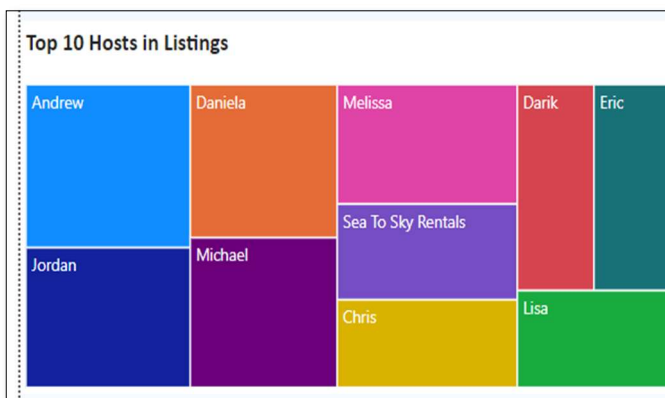
This shows the total number of bookings made for Airbnb homes during the specified time frame.

3. Cancellation Rate (%):

- A **KPI card** showing the **cancellation rate** for bookings during the selected period: **33.32%**.
- This indicates that approximately 1 in 3 bookings were cancelled.



4. List of the Top 10 Hosts:



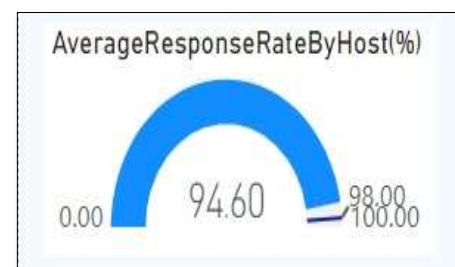
A visual treemap that shows the top 10 hosts about the number of listings they have.

Each rectangle's size indicates how many listings each host has.

Andrew, for instance, has the biggest share, meaning that in comparison to the others, he has the most postings.

5. Average Response Rate by Host (%):

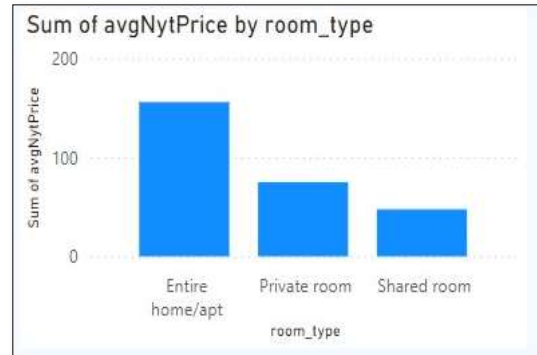
A gauge chart that displays the hosts' average response rate of 94.60%.



This percentage shows how receptive hosts are to questions from visitors.

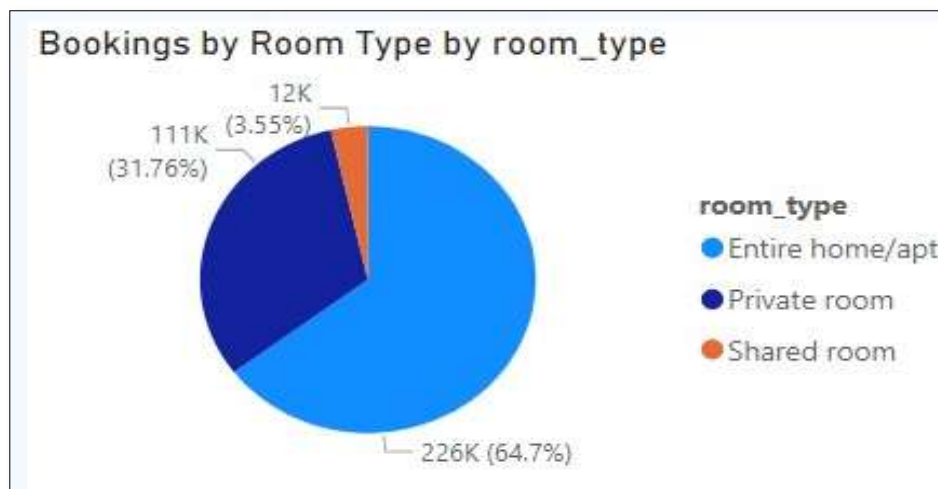
6. Average Nightly Rate Added Up by Room Type:

A bar graph that displays the typical daily cost for each type of room:
At just under \$200, the entire home/apt is the most expensive.
Shared rooms are the least expensive, while both private and shared rooms are more reasonably priced.
This is a summary of the price differences by kind of lodging.

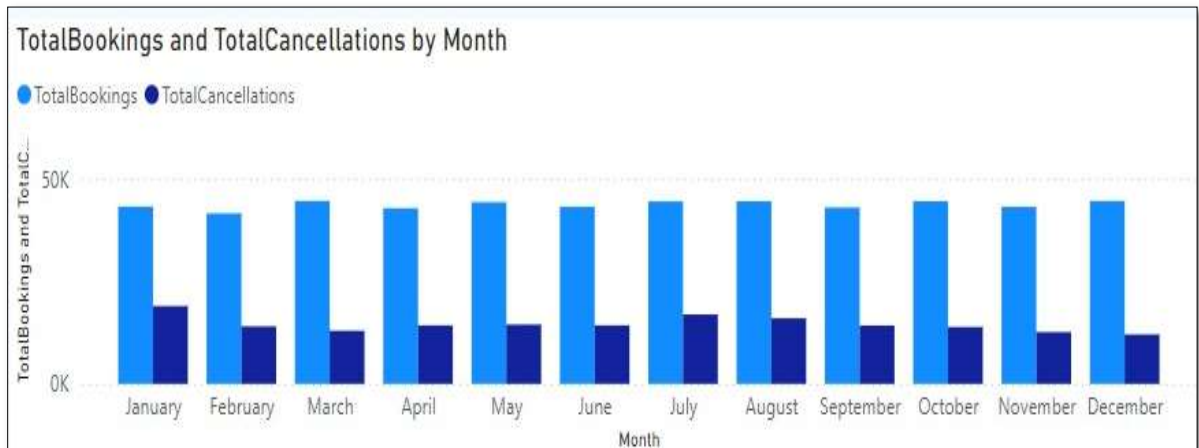


7. Reservations by Type of Room:

A pie chart displaying the total number of reservations divided by kind of room:
64.7% of bookings are for the entire house or apartment.
31.76% of all rooms are private rooms.
There are just 3.55% of bookings that are for shared accommodations.
This graph makes the popularity of each kind of property easier to see.



8. Total Reservations and Total Refunds by Month:



A nested column chart that contrasts the number of reservations and cancellations for each month.

Total bookings are represented by the blue bars, and cancellations are represented by the deeper blue bars.

June and July have large booking volumes and comparatively lower cancellation rates, whereas January displays a lower booking volume and a greater proportion of cancellations.

Seasonal patterns in reservations and cancellations are shown in this graphic.

9. Total Reservations Broken Down by Property Type:

The total number of reservations is broken down by kind of property in this pie chart.

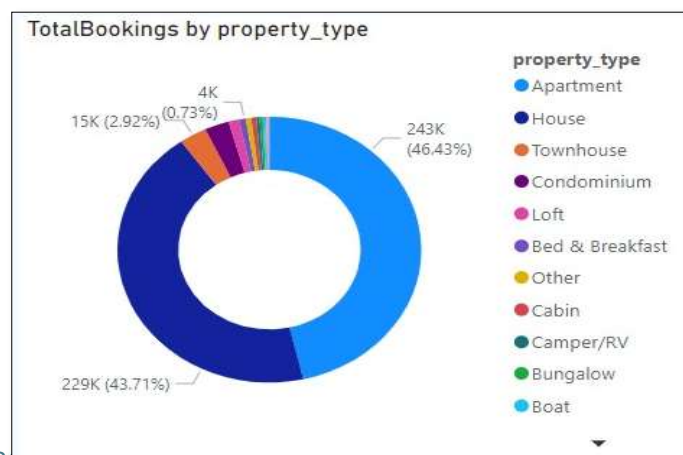
The portions of the pie display the number of reservations made for each category:

With 46.43% of all bookings, apartments make up the greatest chunk.

House follows closely with 43.71% (229K reservations).

The balance is made up of smaller property types such as condominiums (4K bookings, 0.73%) and townhouses (15K bookings, 2.92%).

This graphic presents a clear comparison of the most popular property kinds among visitors.

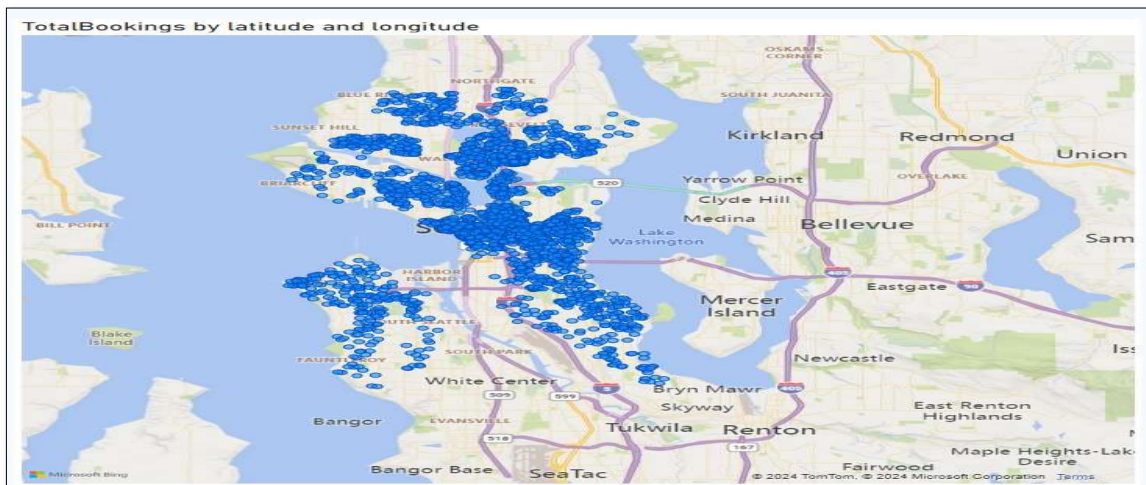


10. Total Bookings by Latitude and Longitude:

The geographic distribution of all reservations made in Seattle and the surrounding areas is depicted using latitude and longitude information in this map representation.

A property with reservations is shown by each blue dot on the map. The density of reservations in particular places is shown by the concentration of dots.

The chart shows that a large concentration of bookings is found in the central Seattle area and waterfront communities like Ballard, Queen Anne, and Capitol Hill. Understanding geographic trends in booking activity can be facilitated by using the visualization to pinpoint hotspots where the majority of Airbnb bookings occur.

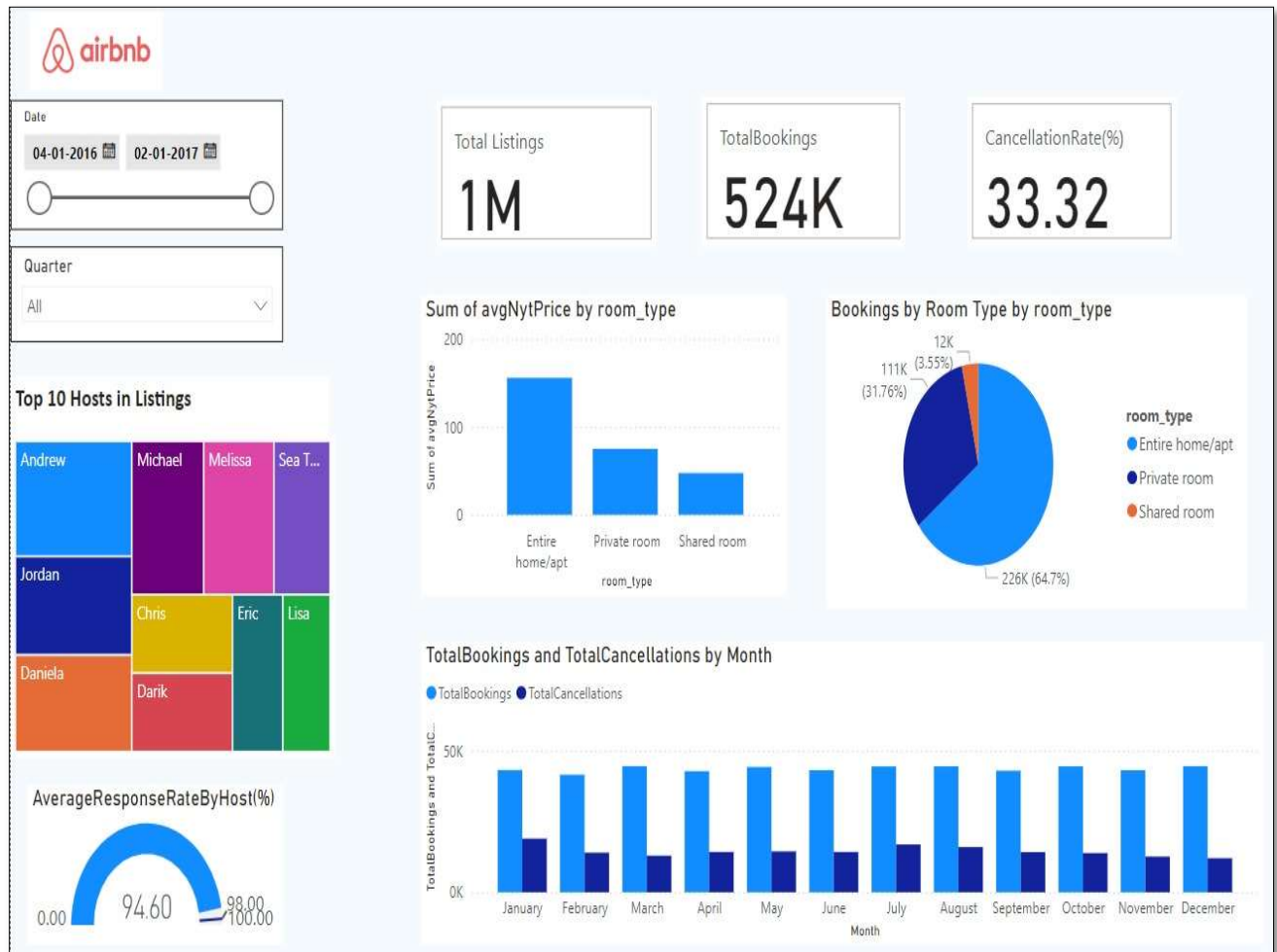


CONCLUSION

According to the Airbnb data, reservations are primarily made for apartments and houses, particularly in central Seattle, where location is very important. Even though they are more expensive, guests prefer full homes or apartments than shared rooms. The highest-ranking hosts are very responsive, which contributes to the retention of booking rates; nonetheless, the notable (33.32%) cancellation rate indicates that more stringent cancellation policies are required. All things considered, the secret to maximum success in the Airbnb market is to optimize property type, location, and host interaction.

Appendix

Dashboard 1



Dashboard 2

